

Learning Objective 3.5: I can implement beam steering on the ADALM-PHASER and verify the steered pattern against theory.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Building the phase table for a new angle.** Your HB100 locks at 10.525 GHz and the PHASER array has $N = 8$ elements on a $d = 14$ mm pitch. You plan to steer to $\theta_0 = 20^\circ$.

- (a) Find the wavelength and the electrical spacing d/λ .

$\lambda =$

$d/\lambda =$

- (b) Compute the progressive element phase $\Delta\phi$ required to steer the beam to 20° .

$\Delta\phi =$

- (c) Complete the table of phases the beamformer will apply, wrapping each value into 0° to 360° . State which element is the first to wrap.

Element	1	2	3	4	5	6	7	8
Applied phase								

- (d) A classmate says the wrapped table cannot produce the same beam as the unwrapped ramp, because element 7 now leads element 1 by only 2.9° instead of lagging a full turn behind. In one or two sentences, explain why the two tables give identical patterns.

2. **Reading a measured sweep.** A cadet places the HB100 at $+30.0^\circ$ on the protractor arc, hunts for the peak in the Beam Steering box, and then runs a full sweep on the Rectangular tab. The trace peaks at a commanded angle of 28.1° , its half-power crossings are at 20.5° and 35.8° , the peak sits 0.7 dB below the frozen boresight trace, and the first sidelobes read -11.8 dBc. The steer resolution is the default 2.8125° .

- (a) Find the measured half-power beamwidth and compare it with the value predicted from the 13.1° boresight beamwidth.

$\theta_{\text{HP}} =$

predicted =

- (b) Express the difference between the physical angle and the commanded angle at the peak as a fraction of one sweep grid step.

offset =

- (c) The cadet reports the array as broken because the beam did not go to 30° . In two or three sentences, say whether the data support that conclusion.

- (d) The peak fell 0.7 dB when the beam was steered to 30° , while the projected-aperture rule predicts a drop of $10\log_{10}(\cos(30^\circ)) = -0.6$ dB instead. Give the most likely reason the measured loss is the larger of the two, and state whether you would expect the gap to grow or shrink at 50° .

3. **Bounding the error sources.** You need to state an uncertainty for the peak angle you measured. Work with the default 2.8125° steer resolution, a source range of 1.00 m, and an HB100 nominal frequency of 10.525 GHz.

- (a) From the sweep grid alone, give the worst-case error in the peak angle read off the trace, and the RMS error if the true peak is equally likely to fall anywhere between two grid points.

worst case =

RMS =

- (b) The phases were computed for 10.525 GHz and applied for $\theta_0 = 45^\circ$. The module then drifts to 10.325 GHz. Find the angle the beam points at, and the resulting error.

$\theta =$

$\Delta\theta =$

- (c) How far along the arc must the tripod be misplaced to introduce a 1.0° error in the reference angle?

$s =$

- (d) Multipath in the classroom adds roughly 1° of peak-angle wander. Combine all four contributions in quadrature, then say which one you could reduce from the GUI and how, and which one you could not.

4. **Working backwards from the phase read-back.** A lab partner recorded the Phase Control values below but forgot which steer angle was commanded. The HB100 was at 10.525 GHz.

Element	1	2	3	4	5	6	7	8
Applied phase	0.0°	125.0°	250.1°	15.1°	140.1°	265.2°	30.2°	155.2°

- (a) Recover the progressive phase $\Delta\phi$ from the table.

$$\Delta\phi = \boxed{}$$

- (b) Find the commanded steer angle.

$$\theta_0 = \boxed{}$$

- (c) The same read-back is also consistent with $\Delta\phi = 125.0^\circ + 360^\circ = 485.0^\circ$. Explain in one or two sentences why that second solution can be discarded for this array.

- (d) The partner later finds the HB100 was running at 10.300 GHz while these phases were applied. Where did the beam point?

$$\theta_0 = \boxed{}$$

5. **Two frozen traces.** A group freezes the boresight sweep, then steers to 45° and sweeps again. The boresight trace has a half-power width of 13.1° and defines the 0 dB reference. The steered trace peaks at 45.0° , is 18.9° wide at half power, and its peak is 1.7 dB below the boresight peak.

- (a) Predict the half-power width at 45° and compare it with the measurement.

$\theta_{\text{HP}} =$

- (b) Predict the peak drop from the projected-aperture rule and compare it with the measured 1.7 dB drop.

$\Delta G =$

- (c) In one or two sentences, name the single geometric fact behind both the wider beam and the lower peak.

- (d) A radar specification calls for a 13° beam maintained out to 45° of scan. How many elements would this array need at the same 14 mm spacing, and state the cost of that choice in one phrase.

$N =$

Documentation: _____